

Adinkra Visualization Suite

A research and validation précis

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The maintained browser suite comprises five interfaces for the length-16 doubly-even code catalog and associated Garden representations. The principal explorer displays all 145 permutation-equivalence classes of positive-dimensional codes, their calculated invariants, and a codeword-to-coordinate incidence graph. A catalog-wide three-dimensional viewer covers all 145 code classes and 5,128 pairs consisting of a code class and a dashing cohomology class. A fly-through viewer presents four selected colored and dashed graphs, and a separate atlas presents the 512 $k = 8$ Garden representations. A gallery displays all 42 checked-in analytical and graph figures generated by seven programs. These products display catalogued and calculated data. They do not establish dimensional enhancement or off-shell supersymmetry.

Scientific context

Adinkra chromotopologies arise from quotients of an N -cube by doubly-even binary codes, with dashings and rankings supplying additional representation data [1, 2, 3, 4]. The 145 catalog entries classify code classes and their associated chromotopologies. They are not 145 complete Adinkras, which also require dashing and ranking data. The visualization suite presents the computational catalog used in the current $N = 16$ program, whose physics motivation includes the off-shell four-dimensional, $N = 4$ Maxwell problem [5].

The visualizations have two purposes:

1. make the discrete catalog and its structural invariants inspectable;
2. present calculated Gadget, spectral, clustering, and graph quantities without requiring a reader to reconstruct them from raw arrays.

The visualizations read files produced by the catalog and representation calculations. They do not recompute the Garden algebra in the browser.

Primary browser explorer

The main application is `visualizer/index.html`. It is a static browser application built with version 7 of the Data-Driven Documents library [6] and loads `adinkra_codes_n16.json`. It provides:

- filtering by code dimension $k = 1, \dots, 8$;
- filters for self-duality, self-orthogonality, and indecomposability;
- sorting by dimension, minimum distance, codeword count, zero columns, automorphism-group size, or catalog index;
- an overview scatter plot with selectable vertical metrics;
- live summaries of the filtered catalog, including the k distribution, decomposability counts, and a zero-column heatmap;
- a detail panel for the generator matrix, weight enumerator, column profile, minimum distance, automorphism-group size, and structural badges;

- a bipartite incidence graph connecting nonzero codewords to the 16 coordinate positions in their support.

The incidence graph visualizes binary-code support. It is distinct from the colored and dashed boson-to-fermion Adinkra graph.

Additional browser interfaces

Artifact	Purpose
<code>visualizer/adinkra_catalog_3d.html</code>	Three.js viewer for a user-selected member of all 145 $N = 16$ code classes and all 5,128 pairs consisting of a code class and a dashing cohomology class, with dashing, generator, edge-budget, and node-inspection controls.
<code>visualizer/adinkra_interactive.html</code>	Three.js fly-through viewer for four selected colored and dashed graphs at $N = 4$, $N = 8$, and the two $N = 16$ self-dual topologies, with free navigation and node inspection.
<code>visualizer/k8_gadget_atlas.html</code>	Plotly interface for the 512 catalogued $k = 8$ representations, with block, Gadget, spectral, and strong-edge views.
<code>visualizer/adinkra_gallery.html</code>	Gallery containing all 42 checked-in analytical and graph figures, grouped by subject.

The seven programs under `viz/` generated 42 checked-in figures: 10 three-dimensional graph renders, 14 Gadget summaries, four spectral or rank-nullity plots, four clustering or code-comparison views, three diffusion-map views, and seven catalog or scaling dashboard panels. These are static outputs, presented together in the gallery.

The Gadget views use the holoraumy-derived comparison developed in the Adinkra literature [7]. Their role here is exploratory comparison and regression inspection, not proof of higher-dimensional liftability.

Data provenance and validation

The principal input declares $n = 16$ and contains 145 permutation-equivalence classes of positive-dimensional doubly-even codes [1]. The distribution by code dimension is

k	1	2	3	4	5	6	7	8
classes	4	10	23	38	36	23	9	2

Every record supplies the fields consumed by the browser panels: generators, codewords, weight distribution, column profile, minimum distance, decomposability, duality properties, zero-column count, and automorphism-group size. The catalog contains two self-dual classes and 104 classes marked indecomposable.

The computational backend has tests for code construction, canonicalization, chromotopology formation, dashing classes, rankings, Garden relations, and known low- N controls. The browser views themselves do not have an automated end-to-end or visual-regression suite.

The catalog exporter writes one compact dashing asset per class. Its manifest contains 145 assets and 5,128 dashing classes. The JavaScript validation constructs sample topologies from four strata and checks the odd-face condition for all 256 dashings of catalog class 75.

Reproduction

Repository: github.com/p1p3dream/adinkra-codespace. From the repository root:

```
cargo run --release -- export-3d-assets
node scripts/test_adinkra_catalog_3d.mjs
python3 -m http.server 8000
```

Then open <http://localhost:8000/visualizer/>. The catalog-wide three-dimensional viewer is at [the catalog 3D viewer](#). The four-example fly-through is at [the interactive navigator](#), and the $k = 8$ interface is at [the Gadget atlas](#). The checked-in figures are at [the figure gallery](#). Serving from the repository root matters because the explorer fetches the catalog from its parent directory. The browser pages load the Data-Driven Documents, Three.js, or Plotly libraries from public content-delivery networks, so they require network access unless those dependencies are vendored.

The tested command-line pipeline performs the representation calculations and catalog construction. The programs under `viz/` generate figures. Separate programs under `scripts/` perform exploratory searches and independent algebraic audits. The rendering dependencies are pinned in `viz/requirements.txt`. The script `scripts/generate_all_visualizations.sh` regenerates and validates the retained visualization products.

Limitations and appropriate use

- The main explorer covers code classes, not every dashing, ranking, or supersymmetry transformation matrix.
- The catalog three-dimensional viewer supports every class and dashing. Its default edge budget limits displayed edges for large low- k graphs. Full display remains available. Dense low- k graphs remain difficult to interpret.
- The interactive fly-through contains four selected graphs. It is not a second catalog-wide viewer.
- The $k = 8$ interface displays checked-in results. It does not calculate arbitrary Gadget values in the browser.
- Checked-in JavaScript Object Notation files and figures are generated artifacts. The repository provides one command that regenerates the catalog assets and all 42 figures, then validates their counts.
- None of the views establishes dimensional enhancement, Lorentz covariance, closure modulo gauge transformations, or the existence of a finite off-shell supermultiplet.

The visualizations provide access to data produced by the tested catalog and representation pipeline. They are not substitutes for the algebraic verification.

Computational artifacts

Path	Role
<code>adinkra_codes_n16.json</code>	Length-16 code catalog consumed by the primary explorer.
<code>visualizer/adinkra_graph_data.json</code>	Data for the four selected colored and dashed graphs in the interactive fly-through.
<code>visualizer/adinkra_dashing/</code>	Compact dashing descriptors and coverage manifest for the catalog three-dimensional viewer.
<code>visualizer/k8_gadget_atlas_stats.json</code>	Summary statistics and thresholds for the $k = 8$ atlas.
<code>visualizer/adinkra_gallery.html</code>	Subject-grouped index of all 42 checked-in figures.
<code>visualizer/*.js</code>	Browser-panel implementation for overview, statistics, graph, and detail views.
<code>viz/</code>	Static figures and their rendering programs.
<code>viz/requirements.txt</code>	Pinned dependencies for the figure generators.
<code>scripts/generate_all_visualizations.sh</code>	Full visualization regeneration and count validation.

References

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